Derivation of CC Gradient in Operator Form

$$\bar{H} = e^{-T} H e^{T} \qquad E = \langle 0|\bar{H}|0\rangle \qquad 0 = \langle \phi|\bar{H}|0\rangle$$

$$\frac{\partial \bar{H}}{\partial \alpha} = \left[\bar{H}, \frac{\partial T}{\partial \alpha}\right] + e^{-T}\frac{\partial H}{\partial \alpha}e^{T} = [\bar{H}, T^{\alpha}] + \bar{H}^{\alpha} \qquad \underline{1} = \sum_{p} |p\rangle\langle p|$$

$$\frac{\partial E}{\partial \alpha} = \langle 0|[\bar{H}, T^{\alpha}]|0\rangle + \langle 0|\bar{H}^{\alpha}|0\rangle = \langle 0|\bar{H}T^{\alpha}|0\rangle - \langle 0|T^{\alpha}\bar{H}|0\rangle + \langle 0|\bar{H}^{\alpha}|0\rangle$$

$$= \sum_{p}\langle 0|\bar{H}|p\rangle\langle p|T^{\alpha}|0\rangle - \sum_{p}\langle 0|T^{\alpha}|p\rangle\langle p|\bar{H}|0\rangle + \langle 0|\bar{H}^{\alpha}|0\rangle$$

$$\boxed{\frac{\partial E}{\partial \alpha} = \sum_{\phi}\langle 0|\bar{H}|\phi\rangle\langle \phi|T^{\alpha}|0\rangle + \langle 0|\bar{H}^{\alpha}|0\rangle}$$

$$0 = \langle \phi|\bar{H}|0\rangle$$

$$0 = \langle \phi|\bar{H}T^{\alpha}|0\rangle - \langle \phi|T^{\alpha}\bar{H}|0\rangle + \langle \phi|\bar{H}^{\alpha}|0\rangle$$

$$= \sum_{p}\langle \phi|\bar{H}|p\rangle\langle p|T^{\alpha}|0\rangle - \sum_{p}\langle \phi|T^{\alpha}|p\rangle\langle p|\bar{H}|0\rangle + \langle \phi|\bar{H}^{\alpha}|0\rangle$$

$$= \sum_{\phi'}\langle \phi|\bar{H}|\phi'\rangle\langle \phi'|T^{\alpha}|0\rangle - \langle \phi|T^{\alpha}|0\rangle E + \langle \phi|\bar{H}^{\alpha}|0\rangle$$

$$= \sum_{\phi'}\left[\langle \phi|\bar{H}|\phi'\rangle - \langle \phi|E|\phi'\rangle\right]\langle \phi'|T^{\alpha}|0\rangle + \langle \phi|\bar{H}^{\alpha}|0\rangle$$

$$0 = \sum_{\phi'}\left[\langle \phi|\bar{H} - E|\phi'\rangle\right]\langle \phi'|T^{\alpha}|0\rangle + \langle \phi|\bar{H}^{\alpha}|0\rangle$$

$$\sum_{\phi'}\left[\langle \phi|\bar{H} - E|\phi'\rangle\right]\langle \phi'|T^{\alpha}|0\rangle = -\langle \phi|\bar{H}^{\alpha}|0\rangle$$

$$\langle \phi'|T^{\alpha}|0\rangle = -\sum_{\phi}\left[\langle \phi'|(\bar{H} - E)^{-1}|\phi\rangle\right]\langle \phi|\bar{H}^{\alpha}|0\rangle$$

$$\frac{\partial E}{\partial \alpha} = -\sum_{\phi}\langle 0|\bar{H}|\phi\rangle\sum_{\phi'}\left[\langle \phi|(\bar{H} - E)^{-1}|\phi'\rangle\right]\langle \phi'|\bar{H}^{\alpha}|0\rangle + \langle 0|\bar{H}^{\alpha}|0\rangle$$

$$\boxed{\frac{\partial E}{\partial \alpha} = -\sum_{\phi\phi'}\langle 0|\bar{H}|\phi\rangle\langle \phi|(\bar{H} - E)^{-1}|\phi'\rangle\langle \phi'|\bar{H}^{\alpha}|0\rangle + \langle 0|\bar{H}^{\alpha}|0\rangle}$$

Define $\Lambda = \Lambda_1 + \Lambda_2 + \ldots$ as a deexcitation operator:

$$\frac{\partial E}{\partial \alpha} = \langle 0|\bar{H}^{\alpha}|0\rangle + \langle 0|\Lambda \bar{H}^{\alpha}|0\rangle$$

where $\langle 0|\Lambda|\phi'\rangle \equiv -\sum_{\phi} \langle 0|\bar{H}|\phi\rangle\langle\phi|(\bar{H}-E)^{-1}|\phi'\rangle$

$$\sum_{\phi'} \langle 0|\Lambda|\phi'\rangle\langle\phi'|(\bar{H}-E)|\phi\rangle = -\sum_{\phi}\langle 0|\bar{H}|\phi\rangle\langle\phi|(\bar{H}-E)^{-1}|\phi'\rangle\sum_{\phi'}\langle\phi'|(\bar{H}-E)|\phi\rangle$$

$$\sum_{\phi'} \langle 0|\Lambda|\phi'\rangle\langle\phi'|(\bar{H}-E)|\phi\rangle + \langle 0|\bar{H}|\phi\rangle = 0$$

$$\langle 0|\Lambda \bar{H}|\phi\rangle - E\langle 0|\Lambda|\phi\rangle + \langle 0|\bar{H}|\phi\rangle = 0 \qquad (*)$$

The energy-containing term may be eliminated by:

$$\langle 0|\bar{H}\Lambda|\phi\rangle = \sum_{\phi'}\langle 0|\bar{H}|\phi'\rangle\langle\phi'|\Lambda|\phi\rangle + \langle 0|\bar{H}|0\rangle\langle 0|\Lambda|\phi\rangle$$

$$= \sum_{\phi'}\langle 0|\bar{H}|\phi'\rangle\langle\phi'|\Lambda|\phi\rangle + E\langle 0|\Lambda|\phi\rangle$$

$$\therefore\ -E\langle 0|\Lambda|\phi\rangle = -\langle 0|\bar{H}\Lambda|\phi\rangle + \sum_{\phi'}\langle 0|\bar{H}|\phi'\rangle\langle\phi'|\Lambda|\phi\rangle$$

Why worry about doing this?

Insterting this into the lambda eqn earlier gives:

$$\boxed{\langle 0|[\Lambda,\bar{H}]|\phi\rangle + \sum_{\phi'}\langle 0|\bar{H}|\phi'\rangle\langle\phi'|\Lambda|\phi\rangle + \langle 0|\bar{H}|\phi\rangle = 0}$$

For $T \equiv T_1 + T_2 + T_3$, the lambda equations through triples are:

$$\langle 0|[\Lambda,\bar{H}]|S\rangle + \langle 0|\bar{H}|S\rangle = 0$$

$$\langle 0|[\Lambda,\bar{H}]|D\rangle + \sum_{S}\langle 0|\bar{H}|S\rangle\langle S|\Lambda|D\rangle + \langle 0|\bar{H}|D\rangle = 0$$

$$\langle 0|[\Lambda,\bar{H}]|T\rangle + \sum_{S}\langle 0|\bar{H}|S\rangle\langle S|\Lambda|T\rangle + \sum_{D}\langle 0|\bar{H}|D\rangle\langle D|\Lambda|T\rangle = 0$$

These can be expressed in a simpler form.
Starting from (*) above, and defining $\mathcal{L} \equiv 1 + \Lambda$,

$$\boxed{\langle 0|\mathcal{L}(\bar{H}-E)|\phi\rangle = 0}$$